

## COPPER HARBOR, MI, USA

WMO: 725387

Lat: 47.467N

Lon: 87.883W

Elev: 626

StdP: 14.37

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94899

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -0.8       | 2.4 | -8.8                          | 3.5 | 1.1  | -4.8 | 4.3 | 3.7  | 35.8                  | 15.0 | 32.5 | 15.9 | 9.2                   | 350  | 0.626 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 12.7                   | 80.9            | 67.3 | 77.5 | 66.1 | 74.8 | 64.8 | 70.1                | 77.5 | 68.3 | 75.0 | 66.5 | 72.9 | 8.2                  | 240  |

| Dehumidification DP/MCDB and HR |       |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 67.3                            | 102.7 | 74.6 | 65.5 | 96.3 | 72.7 | 63.7 | 90.4 | 70.8 | 34.5          | 78.0 | 32.9 | 75.0 | 31.5 | 72.9 | 77.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 29.4          | 26.0 | 23.0 | DB                         | -5.0 | 87.6               | 6.4 | 2.5                                                | -9.6  | 89.4       | -13.3 | 90.9       | -17.0 | 92.3       | -21.6 | 94.1 | (4) |
| (5)               |      |      | WB                         | -5.7 | 74.3               | 6.3 | 1.6                                                | -10.2 | 75.4       | -13.9 | 76.3       | -17.4 | 77.2       | -22.0 | 78.4 | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |                                                                                  | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |      |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                                  |                                                                                  | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |      |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg                                                                            | 42.3   | 19.7  | 19.4  | 28.0  | 37.2  | 48.2  | 57.2  | 64.4  | 65.2  | 58.8  | 46.6  | 35.9  | 25.3 |      |
|                                                                                  | DBStd                                                                            | 18.06  | 9.21  | 9.62  | 10.44 | 8.04  | 7.58  | 6.65  | 6.38  | 5.66  | 7.65  | 7.34  | 8.28  | 8.08 |      |
|                                                                                  | HDD50                                                                            | 4378   | 938   | 857   | 684   | 393   | 126   | 12    | 0     | 0     | 13    | 160   | 429   | 766  |      |
|                                                                                  | HDD65                                                                            | 8483   | 1403  | 1277  | 1146  | 835   | 522   | 249   | 92    | 69    | 213   | 572   | 874   | 1231 |      |
|                                                                                  | CDD50                                                                            | 1562   | 0     | 0     | 3     | 7     | 71    | 227   | 447   | 470   | 278   | 54    | 5     | 0    |      |
|                                                                                  | CDD65                                                                            | 192    | 0     | 0     | 0     | 0     | 2     | 14    | 73    | 74    | 28    | 1     | 0     | 0    |      |
|                                                                                  | CDH74                                                                            | 769    | 0     | 0     | 0     | 0     | 30    | 95    | 281   | 273   | 85    | 5     | 0     | 0    |      |
|                                                                                  | CDH80                                                                            | 122    | 0     | 0     | 0     | 0     | 5     | 18    | 41    | 47    | 11    | 0     | 0     | 0    |      |
| Wind                                                                             | WSAvg                                                                            | 8.8    | 10.2  | 10.8  | 9.2   | 8.0   | 6.9   | 6.0   | 6.9   | 7.5   | 8.4   | 10.1  | 11.1  | 10.7 |      |
| Precipitation                                                                    | PrecAvg                                                                          | 28.2   | 2.1   | 1.3   | 1.5   | 2.2   | 3.0   | 2.7   | 2.4   | 2.1   | 3.0   | 2.9   | 2.2   | 2.2  |      |
|                                                                                  | PrecMax                                                                          | 39.2   | 5.5   | 2.6   | 3.5   | 4.5   | 6.9   | 6.4   | 6.6   | 6.1   | 8.6   | 8.2   | 5.5   | 4.4  |      |
|                                                                                  | PrecMin                                                                          | 19.3   | 0.5   | 0.4   | 0.0   | 0.6   | 1.3   | 0.5   | 0.9   | 0.4   | 1.0   | 0.7   | 1.0   | 0.2  |      |
|                                                                                  | PrecStd                                                                          | 5.5    | 1.3   | 0.6   | 0.9   | 1.1   | 1.4   | 1.3   | 1.2   | 1.5   | 1.5   | 1.7   | 1.0   | 1.0  |      |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                             | DB     | 41.5  | 45.0  | 61.3  | 65.3  | 78.5  | 82.4  | 85.1  | 85.3  | 81.4  | 72.3  | 59.9  | 44.8 |      |
|                                                                                  |                                                                                  | MCWB   | 37.0  | 38.6  | 51.6  | 51.4  | 63.0  | 67.0  | 70.0  | 70.2  | 67.9  | 61.9  | 51.8  | 40.2 |      |
|                                                                                  | 2%                                                                               | DB     | 37.6  | 39.0  | 50.3  | 57.8  | 70.2  | 76.1  | 80.4  | 80.1  | 76.0  | 65.9  | 54.5  | 40.9 |      |
|                                                                                  |                                                                                  | MCWB   | 34.8  | 34.7  | 43.8  | 46.6  | 59.1  | 63.3  | 67.4  | 66.8  | 65.9  | 56.7  | 47.9  | 37.9 |      |
|                                                                                  | 5%                                                                               | DB     | 35.5  | 35.4  | 45.4  | 53.3  | 64.3  | 72.3  | 77.2  | 76.8  | 72.6  | 61.5  | 50.0  | 38.5 |      |
|                                                                                  |                                                                                  | MCWB   | 33.6  | 32.6  | 39.7  | 43.7  | 54.6  | 61.0  | 66.0  | 66.3  | 64.2  | 55.0  | 44.9  | 36.0 |      |
|                                                                                  | 10%                                                                              | DB     | 33.1  | 32.3  | 41.2  | 48.7  | 59.8  | 68.4  | 74.4  | 73.9  | 69.7  | 57.4  | 46.9  | 36.3 |      |
|                                                                                  |                                                                                  | MCWB   | 30.9  | 29.7  | 36.8  | 41.5  | 51.3  | 59.1  | 64.9  | 65.1  | 62.7  | 52.5  | 42.7  | 34.2 |      |
|                                                                                  | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | 37.5  | 39.5  | 54.3  | 53.8  | 66.4  | 69.6  | 73.0  | 72.5  | 70.3  | 63.0  | 53.0 | 41.8 |
|                                                                                  |                                                                                  |        | MCDB  | 39.7  | 43.4  | 59.4  | 63.3  | 73.7  | 78.4  | 80.8  | 80.5  | 77.1  | 70.2  | 57.7 | 43.5 |
| 2%                                                                               |                                                                                  | WB     | 35.3  | 35.3  | 44.3  | 48.3  | 60.2  | 65.7  | 69.7  | 70.2  | 67.9  | 58.9  | 48.9  | 38.3 |      |
|                                                                                  |                                                                                  | MCDB   | 37.0  | 38.5  | 48.7  | 56.4  | 67.7  | 72.3  | 77.4  | 77.1  | 74.1  | 64.6  | 53.8  | 40.1 |      |
| 5%                                                                               |                                                                                  | WB     | 33.4  | 32.4  | 40.2  | 44.7  | 56.1  | 62.8  | 67.9  | 68.3  | 65.6  | 55.5  | 45.8  | 36.4 |      |
|                                                                                  |                                                                                  | MCDB   | 35.4  | 35.0  | 44.9  | 51.6  | 63.1  | 69.9  | 74.7  | 74.6  | 71.2  | 59.9  | 49.7  | 38.2 |      |
| 10%                                                                              |                                                                                  | WB     | 30.9  | 30.0  | 36.9  | 41.9  | 52.1  | 60.1  | 66.2  | 66.3  | 63.5  | 53.0  | 43.6  | 34.1 |      |
|                                                                                  |                                                                                  | MCDB   | 33.1  | 32.3  | 40.9  | 48.2  | 58.4  | 66.8  | 72.7  | 72.5  | 68.6  | 57.3  | 46.6  | 36.1 |      |
| Mean Daily<br>Temperature<br>Range                                               |                                                                                  | 5% DB  | MDBR  | 9.8   | 11.6  | 13.2  | 12.8  | 15.1  | 16.3  | 14.5  | 12.7  | 12.5  | 9.8   | 9.2  | 9.0  |
|                                                                                  |                                                                                  |        | MCDBR | 12.5  | 16.4  | 18.0  | 20.6  | 23.0  | 22.3  | 17.7  | 16.3  | 16.1  | 15.6  | 12.9 | 10.7 |
|                                                                                  | MCWBR                                                                            |        | 10.4  | 12.8  | 12.6  | 12.6  | 14.0  | 13.2  | 9.9   | 8.8   | 8.7   | 9.9   | 9.0   | 9.3  |      |
|                                                                                  | 5% WB                                                                            | MCDBR  | 12.0  | 14.1  | 16.8  | 19.2  | 22.2  | 20.6  | 16.4  | 14.6  | 14.6  | 13.8  | 11.4  | 9.7  |      |
|                                                                                  |                                                                                  | MCWBR  | 10.3  | 11.6  | 12.4  | 12.7  | 15.2  | 14.1  | 10.6  | 9.0   | 9.4   | 9.7   | 8.8   | 9.0  |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub                                                                             | 0.273  | 0.288 | 0.311 | 0.342 | 0.374 | 0.382 | 0.394 | 0.374 | 0.366 | 0.331 | 0.301 | 0.269 |      |      |
|                                                                                  | taud                                                                             | 2.488  | 2.424 | 2.399 | 2.371 | 2.382 | 2.378 | 2.399 | 2.459 | 2.459 | 2.521 | 2.526 | 2.513 |      |      |
|                                                                                  | Ebn at Noon                                                                      | 264    | 281   | 290   | 289   | 282   | 280   | 274   | 276   | 268   | 262   | 249   | 252   |      |      |
|                                                                                  | Edh at Noon                                                                      | 20     | 26    | 31    | 36    | 37    | 37    | 36    | 32    | 30    | 24    | 19    | 17    |      |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg                                                                           | 285    | 564   | 1053  | 1484  | 1784  | 1954  | 1973  | 1702  | 1216  | 707   | 338   | 210   |      |      |
|                                                                                  | RadStd                                                                           | 24     | 67    | 126   | 171   | 118   | 137   | 96    | 117   | 67    | 87    | 55    | 27    |      |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (21 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page